

Pythagoras' Theorem & Trigonometry

Extra practice test 2 — answer sheet · 27 marks total

Section A — Pythagoras' theorem

A1. $8^2 + 6^2 = 64 + 36 = 100$ (1) distance = $\sqrt{100} = \mathbf{10 \text{ km}}$ (1)

A2. $24^2 + 7^2$ (1) = $576 + 49 = 625$ (1) $\sqrt{625} = \mathbf{25 \text{ km}}$ (1)

A3. $9^2 + 12^2 = 81 + 144 = 225$ (1) $15^2 = 225$ — since these are equal, the triangle is **right-angled** (1)

Section B — SOH CAH TOA basics

B1. The correct inverse function ($\sin^{-1}$, $\cos^{-1}$ or $\tan^{-1}$) depends on which two sides are known — $\sin^{-1}$ only works when the opposite and hypotenuse are known (1)

B2. **TOA**: opposite = adjacent $\times \tan \theta$ (1)

B3. **True** — $\cos 60^\circ = 0.5$ and $\sin 30^\circ = 0.5$ (1)

Section C — Finding missing sides

C1. $\tan 34^\circ = h \div 20$ (1) $h = 20 \times \tan 34^\circ$ (1) = **13.5 m** (1)

C2. $\tan 8^\circ = 1200 \div d \rightarrow d = 1200 \div \tan 8^\circ$ (1) = **8538 m** (1)

C3. $\sin 63^\circ = h \div 10 \rightarrow h = 10 \times \sin 63^\circ$ (1) = **8.9 m** (1)

Section D — Finding missing angles

D1. $\sin \theta = 1.2 \div 5 = 0.24 \rightarrow \theta = \sin^{-1}(0.24)$ (1) = **13.9°** (1)

D2. $\tan \theta = 62 \div 45 = 1.3778$ (1) $\theta = \tan^{-1}(1.3778)$ (1) = **54.0°** (1)

Section E — Applying Pythagoras & trigonometry

E1. Half the base = 5 cm; height² = $13^2 - 5^2 = 169 - 25 = 144$ (1) height = $\sqrt{144} = \mathbf{12 \text{ cm}}$ (1)

E2. (a) distance² = $6^2 - 5.2^2 = 36 - 27.04 = 8.96 \rightarrow \sqrt{8.96} = \mathbf{3.0 \text{ m}}$ (1) (b) $\sin \theta = 5.2 \div 6$ (1) $\theta = \sin^{-1}(5.2 \div 6) = \mathbf{60.1^\circ}$ (1)

Marking note: award method marks for a correctly set-up equation or calculation even if the final answer contains a rounding or arithmetic slip. Accept any correct rounding to 1 decimal place unless a different level of accuracy is specified. Missing units on an otherwise-correct numerical answer should lose the final accuracy mark, not the method mark.